COMPUTATIONAL STUDY OF LUNG LOBAR SLIDING AND THE MECHANICS OF BREATHING

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A. Specific Aims

An understanding of lung mechanics can aid in the development of new lung disease treatment and diagnosis strategies. Important historical examples include the introduction of a positive end expiratory pressure to patients under mechanical ventilation to reduce ventilator induced lung injury and the standardization of spirometry to diagnose diseases such as chronic obstructive pulmonary disorder. One poorly understood aspect of lung mechanics is the sliding of the lung lobes against each other along the lobar fissures. Lobar sliding is particularly interesting because of the large variance between different subjects' fissure anatomy. In addition to variations in lobar fissure orientation and geometry, some subjects have incomplete, missing, or auxiliary lobar fissures which could affect lobar sliding and, as a result, their lung mechanics.

Previous researchers have postulated that lung lobar sliding reduces parenchymal distortion in the lungs, but this hypothesis remains untested. In addition, the existing literature currently lacks accurate visualizations and characterizations of lobar sliding. This knowledge gap limits our intuition on lobar sliding and our ability to formulate specific, testable hypotheses about its effects on lung mechanics. I propose using subject-specific contact mechanics models of the lungs to begin filling these gaps in our knowledge of lobar sliding. The specific aims for this project are:

Specific aim #1 is to develop a subject-specific contact mechanics model of the lungs suitable for simulating lobar sliding and its impact on lung deformations. The model geometry and thoracic cavity motion boundary conditions will be derived from CT scans of the lungs. A friction coefficient parameter on the lobar contact constraints will allow the amount of sliding to be controlled. The numerical implementation of contact constraints will be verified with benchmark problems and the accuracy of the simulated interior displacements validated by comparing to anatomical landmark displacements. Additional work will examine the impact of in/excluding gravity in the model.

Specific aim #2 is to test the hypothesis that lung lobar sliding reduces parenchymal distortion using the contact mechanics model from aim #1. For each subject in the study cohort, lung deformation can be simulated with and without lobar sliding and the resulting parenchymal distortion quantified for both scenarios. Thus, causative relationships between lobar sliding and amount of parenchymal distortion can be delineated without confounding factors. During preliminary studies, paired intra-subject comparisons showed lower parenchymal distortion in simulations where lobar sliding was present. Inter-subject comparisons showed that a greater reduction in distortion occurred between the two simulations when more lobar sliding was present.

Specific aim #3 is to visualize and characterize lobar sliding using the contact mechanics model from aim #1. The sliding velocity between contacting points on different lung lobes can be tracked at each simulation time step. This velocity can be convected to the initial undeformed model geometry and integrated over time to produce a detailed visualization of lobar sliding. Intuitively, these integral curves can be interpreted as the "tire tracks" points on one lobe leave behind on the other during a breathing maneuver. In addition to qualitative analysis, quantitative techniques will be applied to further improve our understanding of lobar sliding kinematics. The sliding velocity vector field can be decomposed into three components describing the relative stretching, rotation, and translation between lobes using Helmholtz-Hodge decomposition. The relative contribution of each component to sliding kinematics will be examined.

B. SIGNIFICANCE

Lung mechanics has played an important role in the development of modern lung disease treatment and diagnosis strategies. For example, mortality rates of patients under mechanical ventilation have been significantly reduced by the adoption of mechanically motivated protective ventilation strategies [1]. Mechanical principles have also been leveraged to standardize the practice of spirometry for diagnosing diseases such as chronic obstructive pulmonary disorder [2]. However, the mechanical metrics used in clinical practice, such as lung compliance or forced expiratory volume, often only provide gross measurements of lung function. This limitation can prevent the early diagnosis of diseases that begin localized to small regions of the lung or lead to difficulty distinguishing between potential causes of declining lung function.

Increasingly sophisticated medical image acquisition and analysis tools such as polarized gas MR [3] and CT image registration [4] allow for development of localized mechanical metrics. Although promising, implementation of localized metrics for clinical applications presents challenges. Lung ventilation and deformation exhibits significant spatial heterogeneity in both normal [5] and diseased [6,7] lungs. As a result of heterogeneity, the value of local metrics may provide limited information about the health of the subject on their own and require additional interpretation based on their location and distribution in the lung. Further complications arise from the fact that changes in body position (such as from supine to prone) have been identified with changes in the distribution of lung ventilation and deformation due to gravitational loads and repositioning of the lungs' surroundings [8–10]. Interpretation of local mechanical metrics thus requires a more rigorous understanding of the various physiological aspects that contribute to lung mechanics and the interactions between them.

One often-overlooked aspect of lung mechanics is the sliding motion between lobes of the lung. The lungs are typically divided into five lobes (see figure 1) that are separated from each other by thin layers of pleura along the lobar fissures. This separation allows the lobes to slide against each other during breathing as has been observed in dog lungs using radiopaque markers [11,12] and human lungs using hyperpolarized gas MR [3] and deformable image registration of CT scans [13,14]. However, lobe and fissure anatomy can vary significantly between subjects. In addition to variations in the morphology of lung lobes and fissures [15,16], many subjects exhibit incomplete, missing, or auxiliary fissures [17,18]. These anatomical differences may alter the mechanics of the lungs through the intermediate mechanism of lobar sliding.

Descriptions of lobar sliding kinematics in the literature are limited. Hubmayr et al. and Rodarte et al. quantified lobar sliding in dog lungs by fitting linear transformations to the displacements of parenchymal markers in each lobe and reporting the differences in rotations between lobes [11,12]. They found that magnitude and direction of rotation differences varied with the positioning of the dogs (supine vs. prone) and whether the dogs were under mechanical ventilation or breathing spontaneously. However, this approach to quantifying lobar sliding may report misleading results since a single linear transformation does not adequately capture the nonuniform and nonlinear deformations undergone by each lung lobe during breathing. Amelon et al. [13] found that lobar sliding created a shearing artifact in deformable image registration displacement fields that could be used as a proxy for the amount of sliding at that location. Using this method, they observed that the amount of sliding increased with distance from the mediastinum (the space separating the two lungs). They also found that relatively little sliding occurred on the fissure between the right middle lobe and the right upper lobe compared to the other fissures. They noted that these observations were consistent with the description of lobar sliding as rotations about the hilum. The literature currently lacks visualizations or descriptions of lobar sliding that incorporate the full kinematics of lobar sliding rather than focusing on one aspect (such as rotations or amount of sliding). The lack of an accurate picture of lobar sliding can limit the ability of researchers to formulate specific, testable hypotheses about lobar sliding and its impact on lung mechanics.

Little is known about the mechanical role of lobar sliding. While studying regional ventilation and lobar kinematics in dogs, Hubmayr et al. [11] and Rodarte et al. [12] postulated that lobar sliding helps the lungs reduce parenchymal distortion and stress concentrations by providing them with additional degrees of freedom to adapt to changes in thoracic cavity shape during breathing and between postural changes. However, rigorously testing these theories can be difficult. Observational studies using imaging techniques such as deformable image registration to compare deformation of the lungs in subjects with and without lobar fissures will have difficulty delineating the role of lobar sliding in the presence of confounding variables such as differences between subjects' depth of inspiration, thoracic cavity motion, and tissue properties. An in-vivo experimental intervention that could disallow sliding between lung lobes for paired comparison without otherwise altering the lungs' deformation capabilities would be difficult to implement even in animal studies. Alternatively, the role of lobar sliding can be delineated by utilizing computational models to simulate a breathing maneuver with and without lobar sliding. This approach allows for precise control of confounding factors by incorporating lobar sliding as a parameter that can be controlled independently of other model inputs.

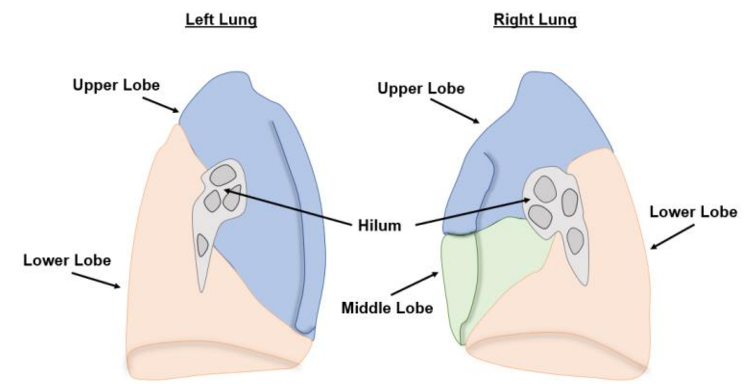

Figure 1: Typical lobar anatomy for each lung as seen from the heart. The left lung is divided into two lobes (upper and lower) whereas the right lung is divided into three (upper, middle, and lower). Lobes are typically separated by a thin layer of pleura along their boundaries called the lobar fissures that allows for sliding between lobes, but these fissures may be missing or incomplete. The bronchi, pulmonary artery, and pulmonary veins enter/exit the lung from the hilum.

A computational experiment requires a model that captures the most important physics associated with lung deformation and lobar sliding in order to deliver reliable results. First, the computational model must be able to capture the deformation kinematics of the lung parenchyma. The finite element (FE) method is particularly well-suited to simulate the deformation of lung parenchyma because it naturally incorporates both the full three-dimensional nature of lung deformation as described in Amelon et al. [5] and the lung tissue interdependence principle described in Mead et al. [19]. The first application of FE modeling in the field of lung mechanics came from West and Matthews with their study on stresses and strains in the lung parenchyma in the presence of gravitational forces [20] and the method has had continued use to study different aspects of lung mechanics such as thoracic cavity shape and motion [10], airway restrictions [21], and the aforementioned gravitational loading [22]. Second, the computational model must be able to simulate lobar sliding. Although there are several different FE studies that incorporate sliding motion between the lungs and the thoracic cavity [23–25] most stop short of incorporating sliding motion between the lobes of the lungs. One notable exception is the FE study of lung lobar sliding by Lee et al. [26]. Lee et al. simulated inflation of the lungs with and without lobar sliding and concluded that lobar sliding produces more uniform stress distributions in the lungs [26]. However, their model used simplified geometries for the lungs (a paraboloid) and lobar fissures (conical slices) and simplified thoracic cavity boundary conditions that vertically scaled the thoracic cavity to represent diaphragm motion and circumferentially scale the thoracic cavity to represent rib cage motion. A later study by Minaeizaeim et al. would demonstrate the importance of subject-specific thoracic cavity shape and motion in lung mechanics [10]. There is no computational model of the lungs in the literature that incorporates both lobar sliding and subject-specific anatomy and boundary conditions.

C. Innovation

To address the knowledge gap concerning lobar sliding's kinematics and mechanical role, I propose the following innovations:

1. Subject-specific contact mechanics model of the lungs that independently meshes the lobes to simulate lobar sliding. This innovation will address the simplified geometry and boundary conditions of Lee et al. [26] by incorporating subject-specific lung lobe geometries and thoracic cavity boundary conditions derived from CT scans. Incorporation of a friction coefficient parameter between lobar contact surfaces will allow for control of the amount of sliding and provide a valuable tool for delineating the role of lobar sliding in lung mechanics and tracking the kinematics of lobar sliding.

2. Testing the hypothesis that lung lobar sliding reduces parenchymal distortion in the lungs using computational models. This study will be one of the first to directly examine lobar sliding's effect on lung mechanics and the first to examine lobar sliding's effect on lung deformation. By performing two simulations with the contact mechanics model for each subject: one where lobar sliding is allowed and one where it is not, a paired comparison of the amount of distortion with and without lobar sliding can be performed for each subject. This innovation will help delineate the differences in lung mechanics between subjects with complete lobar fissures and those with incomplete or missing fissures.

3. 3D quantification of lung lobar sliding kinematics. Multiple studies have confirmed the presence of lobar sliding, but there is little knowledge on *how* the lobes slide. To addresses this knowledge gap, the position of points on the surface of one lobe (the moving lobe) as well as the sliding velocity of these points relative to the other lobe (the reference lobe) will be tracked using the contact mechanics model. The integral curves of this velocity convected to the undeformed reference lobe geometry can intuitively be interpreted as the "tire tracks" the moving lobe leaves behind on the reference lobe. This innovation will provide the most comprehensive visualization and quantification of lobar sliding kinematics to date.

4. Quantitative decomposition of sliding velocity into three physically meaningful components. I will introduce a novel use of Helmholtz-Hodge decomposition to decompose the sliding velocity vector field on a lobe into three comprehensive, orthogonal vector fields that each capture the components of the velocity field arising from relative stretch, rotation, and translation between lobes. This decomposition can quantify the relative contribution of each component to better describe sliding motion and the mechanisms that drive it. A potential application of this innovation is quantitative comparisons of the sliding kinematics between different subject groups (e.g. normal vs diseased, complete vs. incomplete fissures).

D. Research Approach

The overall goal of this project is to improve our understanding of the mechanics of lung lobar sliding. Aim 1 is to develop a contact mechanics model suitable for simulating lung lobar sliding and probing its effects on lung deformations. Aim 2 is to use that model to test the hypothesis that lung lobar sliding reduces parenchymal distortion during a breathing maneuver. Aim 3 is to visualize and characterize lobar sliding kinematics using the contact mechanics model. A detailed discussion of the progress achieved towards these aims and the remaining work is described below.

Specific Aim 1 – Development of a contact mechanics model of lung lobar sliding

The goal of this aim is to develop a contact mechanics model of the lungs capable of simulating lobar sliding and implementing it using the open-source FE software FEBio [27]. This goal will be achieved in three steps; two of which have already been completed. First, the numerical methods utilized in the model were verified using a benchmark problem from the contact mechanics literature (*Aim 1a*). Second, a contact mechanics model of the lung with subject-specific geometry and boundary conditions was implemented in FEBio and the numerical parameters associated with the contact constraints were optimized (*Aim 1b*). The results of these first two steps are published in Galloy et al. [28] and summarized here. Future work will involve incorporating gravitational loads into the contact mechanics model to investigate how its in/exclusion impacts model accuracy and predictions (*Aim 1c*).

Specific Aim 1a – Numerical verification of contact mechanics implementation
During a breathing maneuver, the lung parenchyma undergoes large deformations and sliding motion is present along both lobe-lobe and lobe-thoracic cavity interfaces. These complications make the equations governing the mechanics of the lung highly nonlinear. The hollow spheres benchmark problem from the contact mechanics literature [29,30] was replicated to verify this study's implementation of FEBio's sliding elastic contact formulation [29] under such circumstances. The problem consists of two hollow concentric spheres compressed between two rigid plates. The outer sphere (14 mm outer radius, 2 mm thickness) is in direct contact with the two plates whereas the inner sphere (12 mm outer radius, 2 mm thickness) fits snuggly inside the outer sphere. The two spheres' material properties were described by the neo-Hookean strain-energy function:

$$W = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2 \tag{1}$$

Where I_1 is the first invariant of the right Cauchy-Green tensor, J is the determinant of the deformation gradient, and μ and λ are material parameters set to 0.38 MPa and 0.58 MPa respectively. During the simulation, the plates compress the two spheres by displacing 10 mm inward. Eight-fold symmetry was leveraged so that only one octant of the problem needed to be simulated.

The initial configuration for the problem and deformed configurations at different plate displacements during the simulation are shown in figure 2a. A graph of the reaction force on the rigid plates against the plate displacement is shown in figure 2b along with the results Areaias et al. and Zimmerman and Ateshian reported for the same problem [29,30]. The force-displacement curve of the simulation lined

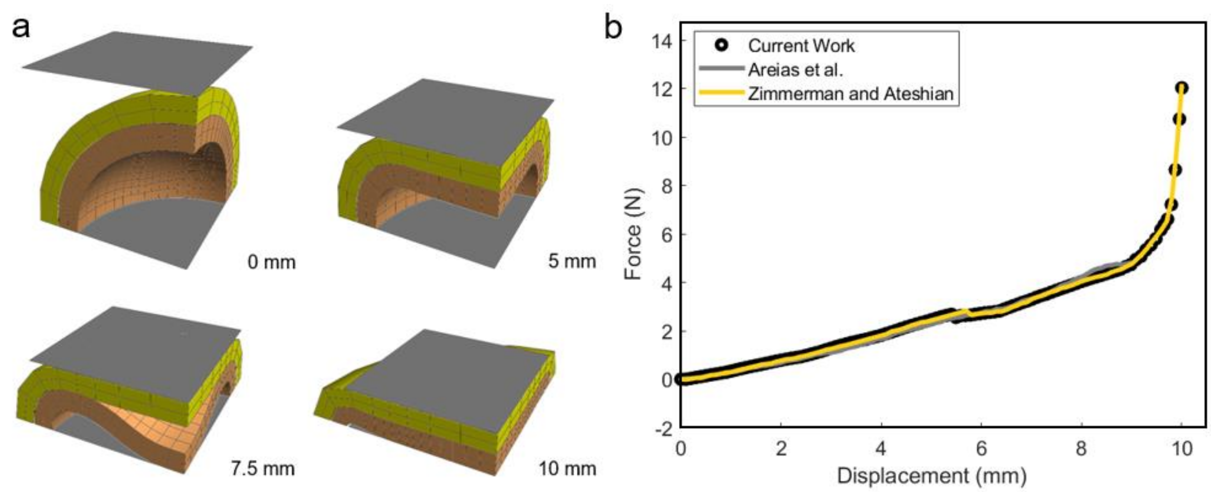

Figure 2: Results for the compression of hollow spheres benchmark problem. (a) The deformed model configuration as the rigid plate lowers. (b) The force vs. displacement curve of the simulation results from the current study agree with previously published results [29,30].

up closely with the ones from the literature. In addition, the inner sphere in the simulation exhibited similar buckling behavior to the previous studies once the plates moved 6 mm inward.

Specific Aim 1b – Optimizing contact mechanics parameters for a subject specific lung model
With the contact mechanics numerical methods verified, the next step was to incorporate the contact mechanics formulation into a subject specific model of the lungs that allowed for lobar sliding.

Model Implementation:
A 4DCT scan of a subject with lung cancer in supine position during tidal breathing was taken using a procedure approved by the University of Iowa Institutional Review Board [4]. The process that produces

FE models from CT images is summarized in figure 3. Although the modeling process is described for the left lung it is readily extended to the right lung (which has three lobes instead of two but is otherwise similar). The left lung and lobes were segmented from the CT scan at end inhale. Volume meshes representing the upper and lower lobes of the left lung were produced from the lobar segmentations. A surface mesh representing the left thoracic cavity used for applying boundary conditions was produced from the left lung segmentation.

When assigning material properties to the lobe volumes several assumptions were made. The lung material properties were assumed to be homogeneous and described by the neo-Hookean strain-energy

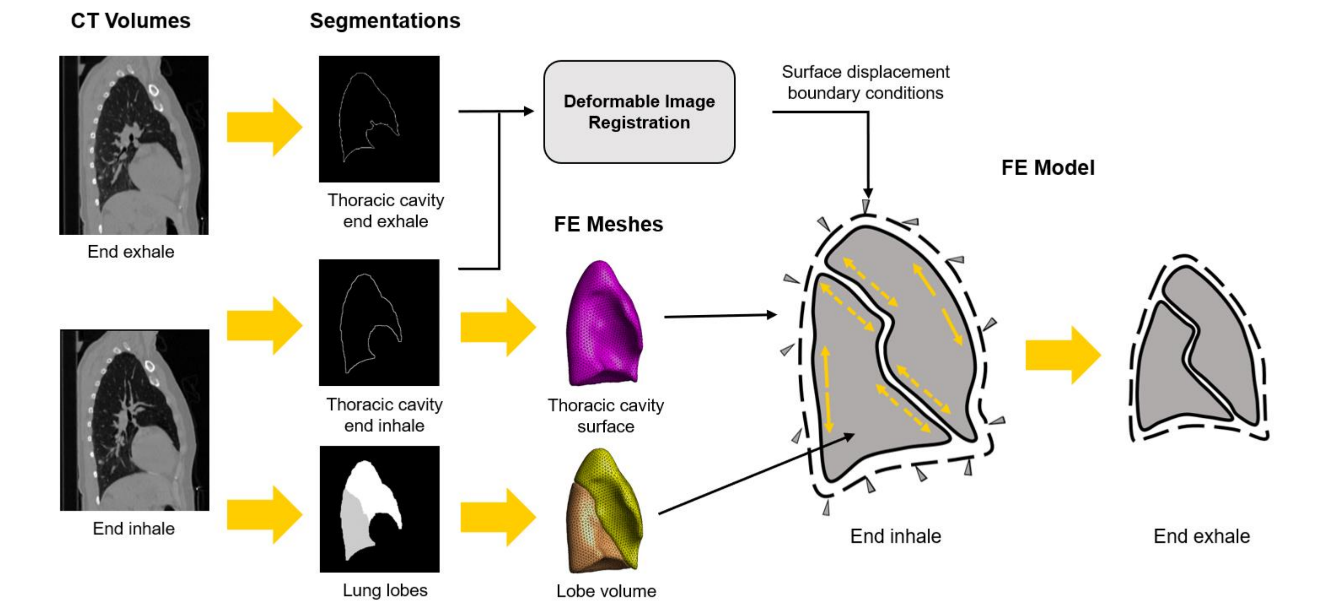

Figure 3: Schematic illustration of pre-processing from CT scans to FE model. The meshes for the lobe volumes and thoracic cavity surface are derived from end inhale segmentations. The displacement field from deformable image registration of the end inhale and end exhale thoracic cavity masks is interpolated onto the thoracic cavity surface and used to apply prescribed displacement boundary conditions. These boundary conditions drive deformation of the lobar volumes through contact constraints between the lobes and the thoracic cavity as well as between the two lobes. A friction coefficient parameter may be used to control the amount of sliding that occurs between the two lobes.

function (equation 1) with parameters μ = 1.15 kPa and λ = 1.73 kPa which have reasonable correspondence with analogous linear elastic properties of lung parenchyma derived from experimental data in dog [31] and rat [32] lungs. The model did not incorporate heterogeneities arising from non-parenchymal lung structures such as the airway tree or cancer tumors. Viscoelastic effects were also not included in the model.

For boundary conditions, nonzero displacements were prescribed to each node of the thoracic cavity surface mesh to simulate motion of the thoracic cavity from end inhale to end exhale. The left lung segmentation mask at end exhale was registered to the mask at end inhale using a sum of squared intensity difference cost function and a cubic B-spline parameterized transformation. Afterward the displacement field from registration was interpolated onto the thoracic cavity surface mesh. The thoracic cavity displacements drove deformation of the lung tissue through frictionless sliding contact constraints between each lobe and the thoracic cavity and between the two lobes. However, unlike the thoracic cavity interfaces a Coulomb friction model was used between the two lobes. This model relates the maximum tangential traction between the lobes to the normal traction between lobes as follows:

$$\mathbf{t}_{t_{max}} = f\mathbf{t}_n \tag{2}$$

Where f is the friction coefficient and $\mathbf{t}_n$ is the normal tractions between lobes. When the friction coefficient is zero, the lobes slide frictionlessly against each other and when the coefficient is nonzero a maximum tangential traction must be overcome for the lobes to slide.

During each simulation, thoracic cavity displacements were incrementally applied over a minimum of ten time steps. The simulations were quasi-static and used a Newton-Rhapson iterative solver. Upon failure of a time step (due to the nonlinear solver exceeding 100 Newton iterations, or inversion of elements) the displacement increment would be reduced, and the time step retried up to five times.

Optimization of contact mechanics parameters:

Contact constraints were enforced using FEBio's sliding elastic contact formulation with augmented Lagrangian constraint enforcement [29] which can be broken into two steps: contact detection and application of surface tractions. During contact detection, points on the primary surface (the lobes) are projected onto the secondary surface (the thoracic cavity) by casting normal rays inward from the primary surface points and finding intersections with the secondary surface. Two passes of contact detection are done for the lobe-lobe interfaces to eliminate bias from choice of primary/secondary surface. The search radius parameter determines the maximum distance rays will be cast to find intersections and must be optimized for the geometry of the problem. A large search radius can result in multiple intersections and ill-defined contact with some geometries (figure 4a), but a small search radius can result in a failure to detect contact (figure 4b).

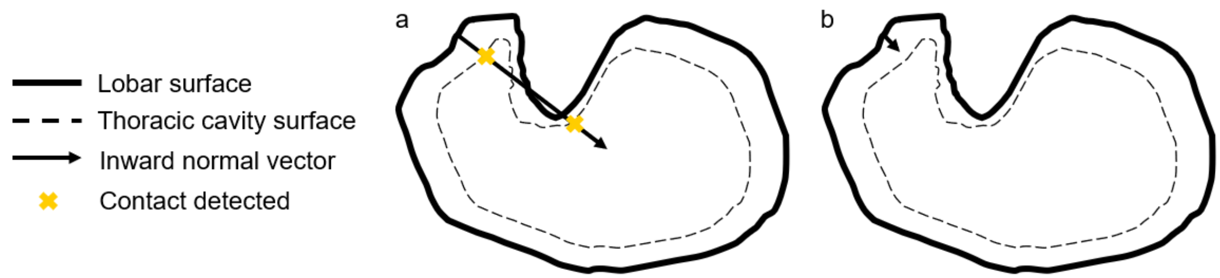

Figure 4: In the above illustrations, contact constraints are being used to enforce contact between the lobar surface and the thoracic cavity surface. (a) Concavity of surfaces may cause a large search radius to identify multiple points of contact. (b) However, if the search radius is too small, a point of contact may not be identified at all.

Once contact between two surfaces is detected, equal and opposite tractions are applied to both surfaces to prevent them from penetrating each other. Normal tractions are iteratively approximated by the following formula (the augmented Lagrangian method):

$$\mathbf{t}_{n,i} = \boldsymbol{\lambda}_{n,i-1} - \epsilon g \boldsymbol{n} \tag{3}$$

Where $\boldsymbol{\lambda}_{n,i-1}$ is the normal traction calculated during the previous augmentation (or the zero vector if this is the first augmentation), g is the penetration gap between the two surfaces, $\boldsymbol{n}$ is the inward unit normal from the primary surface, and ϵ is a penalty parameter. A larger penalty parameter will lead to a greater reduction in penetration during an iteration but can lead to poor numerical conditioning and cause the simulation to fail. Iterative augmentation of the contact tractions ceases when the normalized difference between the current contact tractions and the previous contact tractions is below a threshold called the augmentation tolerance. A lower augmentation tolerance will increase the simulation accuracy but can also significantly increase simulation time.

Thus, three contact mechanics parameters must be optimized for the lung model's stability, speed, and accuracy: the search radius, penalty factor, and augmentation tolerance. Since an appropriate search radius is required for well-defined contact, it was optimized first by running simulations with ten different values of search radius. Simulations with search radii between 0.005 and 0.05 converged. As a result, a search radius of 0.01 was used for all future simulations. Once an appropriate search radius was identified, simulations were run with ten penalty factors ranging from 0.5 to 5.0 in increments of 0.5 and three augmentation tolerances of 0.2, 0.05, and 0.01 for a total of thirty different combinations.

To evaluate the effect of the penalty factor and augmentation tolerance on simulation accuracy, the mean anisotropic deformation index (ADI) was calculated in the model for each set of parameters. ADI is a deformation metric introduced by Amelon et al. [5] to describe the anisotropy of a deformation independent of its volume change. More details on ADI and its relevance to the project are given in *Aim 2*, but here it was simply used to evaluate trends in simulation results with changes in contact mechanics parameters. In theory, the simulation with the largest penalty parameter and the smallest augmentation tolerance should have the most accurate mean ADI but take the most time to run (if it converges).

The results of the penalty factor and augmentation tolerance optimization experiment are summarized in figure 5. All simulations with penalty factors of 3.5, 4.5. and 5.0 did not converge. In general, as contact constraints were more tightly enforced mean ADI increased figure 5a, but this increase was modest. The smallest mean ADI simulated had only an 8.7% difference from the largest mean ADI simulated and all other simulations were within 4% of the largest value. On the other hand, there was a more than five-fold difference in number of iterations between the fastest and slowest simulations figure 5b. The number of iterations required to complete a simulation was most closely tied to the augmentation tolerance, with penalty factor having a less pronounced effect. Based on these results, a penalty factor of 1.5 and augmentation tolerance of 0.2 was selected as the optimal combination with a mean ADI of 0.089 (97.5% largest value) and 74 Newton iterations required to converge.

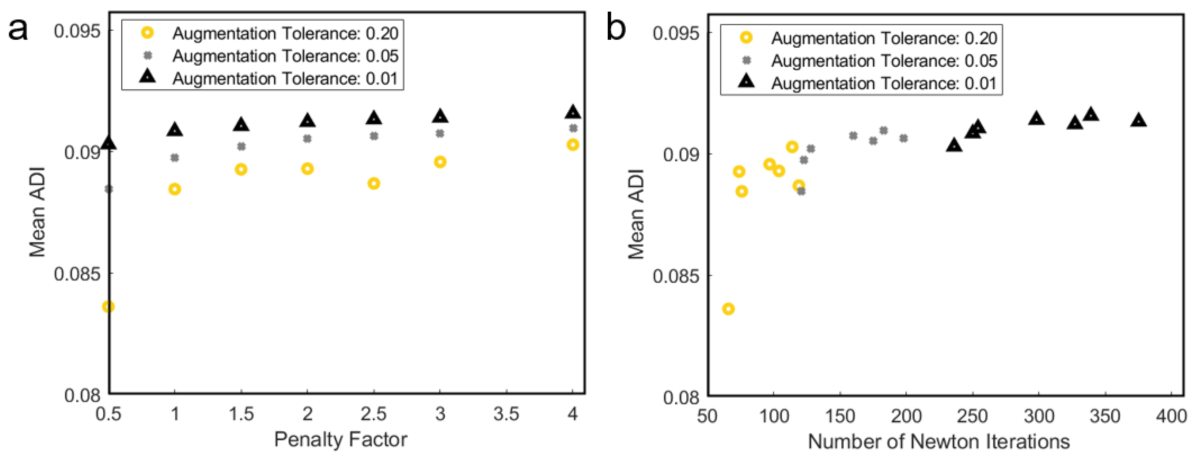

Figure 5: Results of the penalty factor and augmentation tolerance optimization experiment. (a) Mean ADI increased as contact constraints were more tightly enforced (larger penalty factor and smaller augmentation tolerance), but the modest difference indicate the model results are robust to changes in these parameters. (b) On the other hand, simulation time, measured in number of Newton solver iterations, varied considerably with choice of parameters.

With appropriate values for all three contact parameters determined, a final model with search radius 0.01, penalty factor 1.5, and augmentation tolerance 0.2 was examined more closely. Sixty-five anatomical landmarks were manually identified in the end inhale CT image that the model geometry was derived from as well as their corresponding locations in the end exhale image to determine their displacements. These landmark displacements were then compared to the FE model derived displacement field. The model had an average landmark error of 2.3 mm (average landmark displacement of 10.8 mm). These values are comparable to the mean average landmark errors reported by Al Mayah et al. (2.8 mm) [25] and Werner et al. (2.5 mm) [23] for their FE models simulating tidal breathing.

Specific Aim 1c – Investigating the impact of gravitational loading on model results
Gravitational loads are not currently included in the contact mechanics model of the lung. However, it is not clear how (if at all) gravity will affect the hypothesis that lobar sliding reduces parenchymal distortion

(*Aim 2*) or lobar sliding kinematics (*Aim 3*). Gravitational loads on the lungs and surrounding structures are suspected to play a role in the heterogeneity of lung deformation as suggested by previous computational studies [10,20,22]. At functional residual capacity (FRC), the weight of the lungs deflects the lung tissue and there is a gradient in tissue density in the direction of the gravitational force. On the other hand, the lungs are pulled taught at total lung capacity (TLC) resulting in little deflection from the weight of the lungs and the vanishing of the density gradient. This means the lower regions of the lung tissue (with respect to the direction of gravity) are expected to experience larger volume changes and more ventilation between FRC and TLC than the upper regions [10,20,22].

<u>Future work:</u>
Gravitational loading will be implemented into the contact mechanics model to understand its impact on lobar sliding mechanics. Properly incorporating gravity into the contact mechanics model introduces three new challenges to address. First, the current model is displacement-driven and is being used to predict tissue displacements and deformations. As a result, the model is very robust to choice of material constitutive law and material parameters [33,34]. However, this in no longer the case when displacement and force boundary conditions are mixed, so a more accurate constitutive law must be used. The most suitable constitutive law fit to experimental data is the one proposed by Birzle et al. [32]. Second, even at its smallest physiological volumes, the lung is still inflated a significant amount putting it under tension. As discussed above, this state of tension plays an important role in how gravity affects the lung tissue. The model must include a pre-strain to capture this effect, but there is no way to directly measure this pre-strain and assumptions will need to be made. A reasonable assumption could be that TLC (or FRC) is under a uniform volumetric expansion [10,26] with an appropriate volume ratio determined from literature values or CT tissue density analysis. Finally, preliminary attempts at implementing these changes have had issues with simulations failing to converge to a stable solution. Because of these issues, the more robust model from (*Aim 1.2*) without gravitational forces would be preferred if gravity does not have a significant impact on the results of my other aims. The effect of gravity on landmark error and the results reported in other aims will first be examined on the limited number of simulations that have converged to determine if the more complex model is warranted.

Specific Aim 2 – Test the hypothesis that lung lobar sliding reduces parenchymal distortion during breathing
In this aim, the contact mechanics model of the lung developed in Aim 1 will be used to test the hypothesis posed in the lung mechanics literature that lung lobar sliding reduces parenchymal distortion during breathing [11,12]. At this time, this hypothesis has been tested for the left lung using the contact mechanics model developed in *Aim 1.2*. Future work will examine if the inclusion of gravity in the model changes the study outcome and will test the hypothesis in the right lung.

<u>Computational experiment methodology:</u>
4DCT scans from eight subjects with lung cancer during tidal breathing were acquired using a protocol approved by the University of Iowa Institutional Review Board [4]. In addition, six CT scans of healthy subjects during breath holds near FRC and TLC were acquired using a protocol approved by the University of Iowa Institutional Review Board [5]. These two different cohorts were used to examine the effect of lobar sliding on parenchymal distortion at different depths of inspiration. The CT scans for each subject were converted to FE models using the procedure detailed in *Aim 1.2*. For each subject, two simulations were performed: one where the friction coefficient between lobes was set to zero to allow frictionless sliding, and another where a large friction coefficient of 1.5 was set between lobes to practically eliminate sliding.

The anisotropic deformation index (ADI) developed by Amelon et al. [5] was used to quantify the amount of distortion in the simulated lung tissue deformations. ADI is a deformation metric defined on a region of lung tissue by the following equation:

$$\mathrm{ADI} = \sqrt{\left(\frac{\lambda_1 - \lambda_2}{\lambda_2}\right)^2 + \left(\frac{\lambda_2 - \lambda_3}{\lambda_3}\right)^2} \tag{4}$$

Where λ_i are the principal stretch ratios of the deformation in that tissue region sorted from greatest to least. ADI describes the anisotropy of deformation independent of volume change. A low ADI in a region indicates little directional preference in the deformation; that is, a cube of lung parenchymal volume will retain its cube shape as it expands or contracts in volume. A large ADI, on the other hand, indicates strong directional preference in the deformation and thus more geometric distortion in the tissue; that is, a cube of lung parenchymal volume would deform into a slab or a rod shape. The difference in the spatial mean of the ADI between each subject's non-sliding and sliding simulations was determined to quantify the effect of lobar sliding on parenchymal distortion. To determine if the intra subject mean ADI was significantly different between sliding models and non-sliding models, paired two-tailed Wilcoxon signed rank tests with $p < 0.05$ as the criterion for statistical significance were performed.

To quantify the amount of sliding that occurred in each subject's sliding simulation, the FE displacement field results were re-sampled onto a 5 x 5 x 5 mm voxel grid. When performing this interpolation, the corners of voxels at the fissure can be in different lobes. At such voxels, sliding motion between the lobes manifests as shearing of the voxel. Amelon et al. [13] demonstrated that the max shear stretch in an element at the fissure interface can thus serve as a proxy for the amount of sliding at that fissure region. Re-computing the principal stretches on the new grid as above, the max shear stretch is given by:

$$\gamma_{max} = \frac{\lambda_1 - \lambda_3}{2} \tag{5}$$

The mean of the max shear at the fissure was used to quantify the sliding at the fissure that occurred in each subject's FE model.

<u>Results for the left lung:</u>

For all tidal breathing subjects, mean ADI was calculated in the whole left lung and in each individual lobe as a metric to quantify distortion for sliding and non-sliding simulations. Mean ADI throughout the entire left lung was significantly lower in the sliding simulations than the non-sliding ones as seen in figure 6a (5.3% median decrease, $p = 0.008$, $n = 8$). This decrease was more pronounced in the lower lobe (8.0% median decrease, $p = 0.008$) and was insignificant in the upper lobe (1.4% median decrease, $p = 0.078$). For the breath hold subjects, mean ADI throughout the entire left lung was significantly lower in the sliding simulations than the non-sliding ones as seen in figure 6b (3.2% median decrease, $p = 0.031$, $n = 6$). The difference was once again more pronounced in the lower lobe (8.0% median decrease, $p = 0.031$). On the other hand, in the upper lobe some subjects had slightly higher mean ADI in sliding models than non-sliding models while others had slightly lower. Consequently, unlike the lower lobe, mean ADI was not statistically different in the upper lobe (1.8% median increase, $p = 0.063$).

In addition to trends between models from the same subject, trends between subjects were also examined. The mean ADI decrease between sliding and non-sliding simulations for a given subject (this would capture how much allowing sliding reduces parenchymal distortion) was strongly correlated with the mean fissure max shear (i.e. level of lobar sliding) measured in the model that allowed lobar sliding as shown in figure 7 (Pearson's correlation coefficient of 0.95 for tidal breathing, 0.87 for breath holds, and 0.91 for the combined data set).

These intra and inter-subject trends are consistent with the hypothesis that lung lobar sliding reduces parenchymal distortion during breathing. The lower distortion in sliding simulations compared to non-sliding simulations suggests that a subject with incomplete or missing fissures would have greater levels of parenchymal distortion than a subject with complete fissures when all else is equal. In addition, the inter-subject trend displayed in figure 7 suggests that inter-subject variance in the amount (not just presence) of lobar sliding can lead to inter-subject variance in lung distortion and potentially other aspects of lung mechanics.

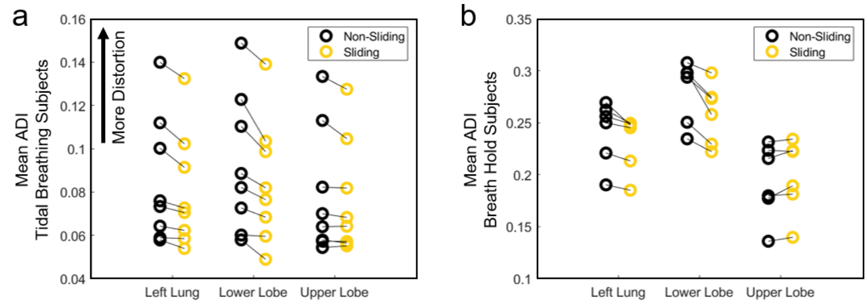

Figure 6: Effect of lobar sliding on parenchymal distortion in each subject for (a) tidal breathing and (b) breath hold cohorts. Spatial mean ADI in the left lung was lower in the sliding model than the non-sliding model for every subject indicating that lobar sliding decreased parenchymal distortion. The reduction in the distortion occurs primarily in the lower lobe.

Future Work:

Since the previous results were obtained using the model from *Aim 1.2* which does not simulate the effects of gravity, they will be re-examined with a smaller set of simulations including gravity to ensure the conclusions drawn from the results still hold. Additionally, the hypothesis will be tested using the same procedure in the right lung.

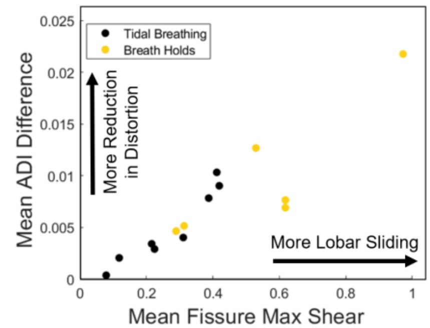

Figure 7: The effect of greater lobar sliding on mean ADI difference between sliding and non-sliding models in each subject. The strong correlation between mean fissure max shear and mean ADI difference suggests that, consistent with the hypothesis, more lobar sliding led to a greater reduction in parenchymal distortion when fissure sliding was permitted.

Specific Aim 3 – Quantitative characterization of lobar sliding

The goal of this aim is to use simulation results from the contact mechanics model (*Aim 1*) to address the literature's lack of an accurate and detailed characterization of lobar sliding kinematics. The first step towards this will be to develop an accurate visualization and quantification of lobar sliding (*Aim 3a*). Next, the use of Helmholtz-Hodge decomposition to break sliding motion into three meaningful components representing relative stretch, rotation, and translation between lobes of the lungs will be investigated (*Aim 3b*). Of note, the techniques here are not exclusive to the computational model and could potentially be applied to CT data with registration derived displacement fields and adequate temporal resolution. The model will be used for this project since it has high temporal resolution displacement fields that best illustrate the utility of these techniques.

Specific Aim 3a – Quantification of lobar sliding kinematics

The goal of this aim is to visualize lobar sliding in a way that captures as much kinematic nuance as possible while still being intuitive to interpret. For a given lobar fissure one lobe will be designated the "moving lobe" while the other will be designated as a "reference lobe". Using the results of the FE simulations, a set of material points on the surface of the moving lobe will be tracked from end inhale to end exhale. At each time step, the moving lobe points will be projected onto the reference lobe to establish a pair of contacting points. From this pairing an incremental sliding displacement vector for each moving point at time step n can be defined as:

$$\Delta \mathbf{s}_n = \boldsymbol{\pi}^{(2)} \left(\mathbf{u}_{n+1}^{(1)} - \mathbf{u}_n^{(1)} - \mathbf{u}_{n+1}^{(2)} + \mathbf{u}_n^{(2)} \right) \tag{6}$$

Where $\mathbf{u}_n$ is the displacement vector at time n, a superscript (1) indicates a value on a moving lobe point, a superscript (2) indicates a value on the corresponding point of contact on the reference lobe, and $\boldsymbol{\pi}^{(2)}$ is the projection operator onto the tangent plane of the reference lobe at the point of contact. If normalized to the duration of the time step, this incremental displacement also serves as an approximation of the sliding velocity. By adding up the magnitudes of each sliding displacement increment, the total sliding distance traversed by the moving lobe point can be computed and mapped across the lobar fissure. In addition to computing a sliding distance, the sliding velocities may be convected to the undeformed reference lobe geometry to produce convected sliding velocity integral curves [35]. These integral curves may intuitively be thought of as the "tracks" the moving lobe leaves behind on the reference lobe during a breathing maneuver. Figure 8 shows an example of both a sliding distance contour plot (figure 8a) and the convected sliding velocity integral curves (figure 8b) with the left lower lobe as the moving surface and the left upper lobe as the reference surface during exhalation from TLC to FRC.

<u>Future work:</u>
This visualization technique will be used to qualitatively describe and identify trends in sliding kinematics for the left lung's lobar fissure and the right lung's three lobar fissures.

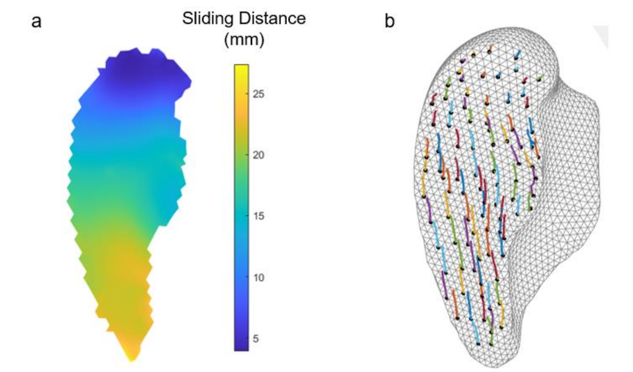

Figure 8: (a) A contour plot of sliding distance along the lobar fissure of the left lung. (b) The colored curves indicate the convected sliding velocity integral curves on the undeformed upper lobe surface (black dots indicate starting point). Intuitively, these curves can be interpreted as the tracks left behind by lower lobe points as they slid across the upper lobe during exhalation from TLC to FRC.

Specific Aim 3b – Component-wise analysis of lobar sliding with Helmholtz-Hodge decomposition
One way to characterize the kinematics of lobar sliding is to decompose this complex motion into simpler components. For example, Hubmayr et al. and Rodarte et al. described lobar sliding in dog lungs by quantifying differences in rotations between lung lobes [11,12]. However, the lung lobes undergo significant nonuniform parenchymal deformation which cannot be described by a single set of rotations. Helmholtz-Hodge decomposition (HHD) is promising for the characterization of sliding kinematics because it can decompose a highly nonuniform vector field on a surface into three component vector fields associated with stretching, rotation, and translation. The goal of this aim is to apply HHD to characterize sliding kinematics with physically meaningful components that still capture the full nonuniformity and complexity of sliding motion.

For a vector field tangent to a surface homeomorphic to an open disc (such as the sliding velocity field derived in *Aim 3a* on a lobar fissure surface), HHD identifies three orthogonal components: the exact component, the coexact component, and the harmonic component [36]. The exact component is curl-free and contains all the divergence of the vector field. For the sliding velocity field, it describes sliding motion arising from one lobe stretching differently than the other. The coexact component is divergence-free and contains all the curl of the vector field. For the sliding velocity field, it describes the sliding motion arising from one lobe rotating about the other. The harmonic component is both curl and divergence-free and describes translational sliding motion. An example HHD of a sliding velocity field on the left lung lobar fissure is shown in figure 9.

As a proof of concept, for each of the six breath hold subjects from *Aim 2* the square magnitude of each HHD component was normalized to the square magnitude of the whole vector field to determine the relative contribution of each component. Figure 10a shows the relative contributions of each component for each subject. On average, 88% of the sliding velocity vector field was captured by the harmonic component, 7.5% by the coexact component, and 4.2% by the exact component. The dominating harmonic component suggests that lobar sliding is mostly translational and has little relative stretching or rotation between lobes. The exact and coexact components had varied contributions to the overall sliding motion ranging from 2.0% to 7.4% and 0.5% to 21.6% respectively.

Figure 10b-c compares the sliding displacement field streamlines of subject 4 who had the largest exact and coexact contributions to subject 6 whose motion was almost entirely harmonic. Subject 4 exhibits an inward spiral towards the top right (corresponding anatomically to an apical lateral location) indicating both rotation and compression of the lower lobe relative to the upper lobe at that location. On the other hand, subject 6's streamlines form a nearly parallel field.

<u>Future work:</u>

The numerical techniques associated with HHD will be investigated to ensure their application to this project is accurately interpreted. Boundary effects can sometimes lead to misleading interpretations of HHD results [36,37]. Furthermore, HHD can only decompose the sliding vector field at one time step, so variations in the HHD components throughout the simulation should be investigated. Afterwards HHD will be used to quantitatively characterize lobar sliding in all fissures of the left and right lungs.

Conclusion:

Understanding the mechanics of breathing can lead to better diagnosis and treatment of lung diseases. Lobar sliding is an often-overlooked aspect of lung mechanics that can have implications for understanding how variations in fissure anatomy such as missing, incomplete, or auxiliary fissures affect lung health. The goal of this project is to study the kinematics and mechanical role of lobar sliding using a contact mechanics model of the lungs. Significant work has already been completed developing the contact mechanics model, delineating the effect of lobar sliding on parenchymal distortion for the left lung, and implementing the primary quantitative techniques that will be used for the project. Future work will focus on applying the techniques described here to both the left and right lungs as well as identifying the impact of gravity on model predictions. Current progress and a timeline for the project's remaining work stratified by aim is provided in table 1.

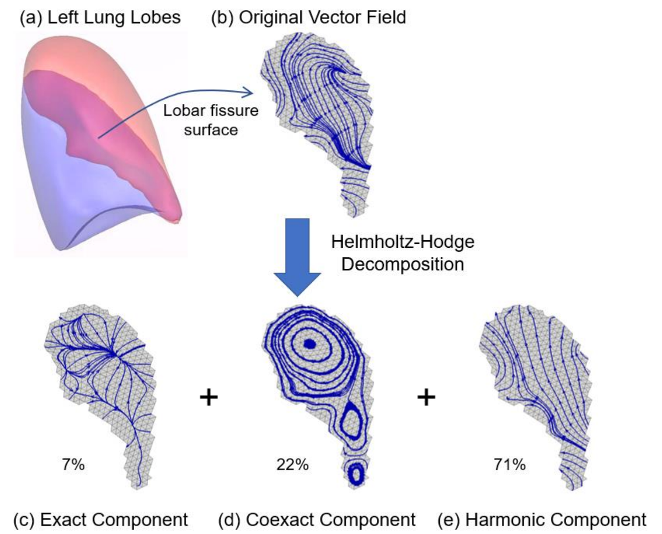

Figure 9: 3D model of the left lung showing the lobar fissure (a) with the streamlines of the sliding displacement vector field (b). The vector field was decomposed into three HHD components: a stretch induced exact component (c), a rotational coexact component (d), and a translational harmonic component (e).

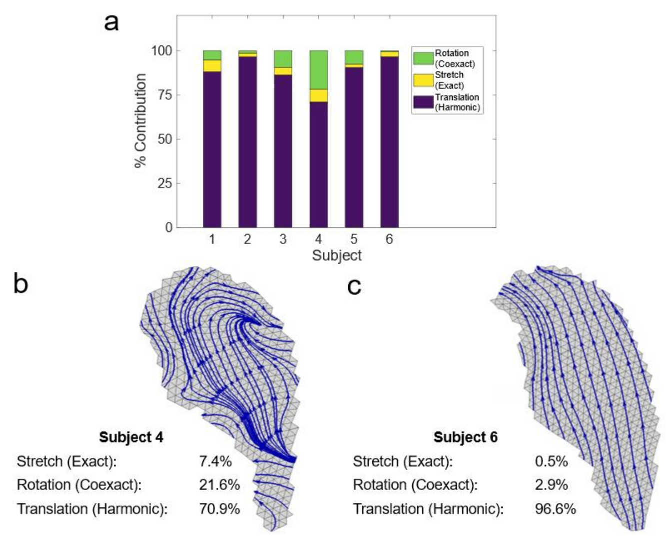

Figure 10: (a) Relative contributions of HHD components in each subject suggest lobar sliding is predominantly translational (harmonic) with lesser contributions from rotational (coexact) and stretch induced (exact) motions. (b) Subject 4's streamlines exhibit an inward spiral that is reflected by larger exact and coexact components. (c) In contrast, subject 6's streamlines are nearly parallel and have small exact and coexact components.

Table 1: Current progress on the project and a timeline for remaining work stratified by aim (dark red) and further stratified by sub-aim (light red) where appropriate. The target for my thesis dissertation defense is October 2023 (dark green).

		PROJECT TIMELINE								
Specific Aim	**Completion**	Feb-23	Mar-23	Apr-23	May-23	Jun-23	Jul-23	Aug-23	Sep-23	Oct-23
Aim 1: Development of a contact mechanics model of lung lobar sliding	**90%**									
Aim 1a: Verify numerical methods with benchmark problems	100%									
Aim 1b: Develop modeling procedure and optimize parameters	100%									
Aim 1c: Investigate the impact of gravity on model	25%									
Aim 2: Test hypothesis that lobar sliding reduces parenchymal distortion	**80%**									
Aim 3: Visualization and characterization of lobar sliding	**20%**									
Aim 3a: Visualization of lobar sliding kinematics	20%									
Aim 3b: Component-wise analysis of lobar sliding	20%									
Thesis Dissertation Defense										

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